2021-Ph-H1-Q23

Section: Electricity

Topic: Capacitors

A 10 µF capacitor is charged to 12 V. What is the energy stored?

Solution:

$$E = \frac{1}{2}CV^2 = \frac{1}{2} \times 10 \times 10^{-6} \times 144 = 0.00072\,\mathrm{J} = 0.72\,\mathrm{mJ}$$

Answer: D

Guidance for Students:

Use the correct formula with units in farads and volts. Energy will come out in joules.

Revision Tip:

Capacitor energy:

$$E = \frac{1}{2}CV^2$$

1 µF = 10^{-6} F